

Page 1 - Motion

Motion describes the change in position of an object with time. Speed, velocity and acceleration are fundamental concepts.

Practice Questions (QE)

1. Define speed and velocity.
2. A car travels 120 km in 2 hours. Find its average speed.

Page 2 - Newton's Laws

Newton's three laws explain the relationship between force and motion.

Practice Questions (QE)

1. State Newton's Second Law.
2. A 5 kg object accelerates at 3 m/s^2 . Find the force.

Page 3 - Work, Energy and Power

Work is done when a force causes displacement. Energy is the capacity to do work.

Practice Questions (QE)

1. Calculate work done by a 20 N force over 5 m.
2. State the law of conservation of energy.

Page 4 - Gravitation

Every object attracts every other object due to gravity.

Practice Questions (QE)

1. State Newton's law of gravitation.
2. Find the weight of a 60 kg person ($g = 9.8 \text{ m/s}^2$).

Page 5 - Pressure and Fluids

Pressure depends on force and area. Liquids transmit pressure in all directions.

Practice Questions (QE)

1. Define pressure.
2. Why do dams have thicker bases?

Page 6 - Heat and Temperature

Heat is energy transferred because of a temperature difference.

Practice Questions (QE)

1. Differentiate heat and temperature.
2. What is specific heat capacity?

Page 7 - Light

Reflection and refraction explain how light behaves.

Practice Questions (QE)

1. State the laws of reflection.
2. Draw a ray diagram for a convex lens.

Page 8 - Electricity

Current, voltage and resistance are related by Ohm's law.

Practice Questions (QE)

1. State Ohm's law.
2. Find the current if $V = 12\text{ V}$ and $R = 4\ \Omega$.

Page 9 - Magnetism

Magnets produce magnetic fields and interact with electric currents.

Practice Questions (QE)

1. What is an electromagnet?
2. State Fleming's Left-Hand Rule.

Page 10 - Modern Physics

Atoms, radioactivity and semiconductors are part of modern physics.

Practice Questions (QE)

1. What is radioactivity?
2. Name two semiconductor devices.